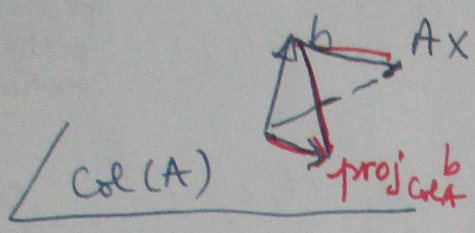
Seminar 10 – CD 312 – Algard.

Aproximarea unui sistem liniar incompatibil

$A \in \mathcal{M}_{m,n}(\mathbb{R})$; $b \in \mathbb{R}^m$; $b \notin Col(A) \Rightarrow \boxed{Ax = b}$ incompat.



? x i. $\|b - Ax\|$ min.

$\min_x \|b - Ax\| = \|b - proj_{Col A} b\|$

$\boxed{Ax = proj_{Col A} b}$ sist compat $\Rightarrow x^*$ pseudosol. sist $Ax = b$.

Descompunerea QR a unei matrici.

$A \in \mathcal{M}_{m,n}(\mathbb{R}) \Rightarrow \underbrace{A}_{m\times n} = \underbrace{Q}_{m\times n} \cdot \underbrace{R}_{n\times n}$

$\{v_1, \dots, v_n\}$ lin ind.

Q ortogonală: $Q \cdot Q^t = I_m$; $Q^t Q = I_n$.

R superior triunghiular nesing.

$$R = \begin{pmatrix} v_1 & v_2 & v_3 & \\ * & * & * & q_1 \\ 0 & * & * & q_2 \\ 0 & 0 & * & \end{pmatrix}$$

$Q = [q_1 | \dots | q_n]$.

$A = [v_1 | \dots | v_n] \Rightarrow \{v_1, \dots, v_n\} \xrightarrow{G\text{-}S} \{q_1, \dots, q_n\}$.

$r_{11} = \langle v_1, q_1 \rangle$; $r_{12} = \langle q_1, v_2 \rangle$.

$r_{21} = \langle q_2, v_1 \rangle = 0$.

$\boxed{Ax = b}$ sist inc ; $\boxed{A = QR}$ $\Rightarrow A^t Ax = A^t b$. sist compat.

$A^t = R^t Q^t$

$A^t A = R^t \underbrace{Q^t Q}_{I} R = R^t R \Rightarrow R^t Rx = R^t Q^t b. \Rightarrow \boxed{Rx = Q^t b.}$

sist compatib

R mat sup triunghi =) $\Rightarrow$ sistem usor de rez.

Ex 1 a) $A = \begin{pmatrix} 1 & 5 \\ 2 & -2 \\ -1 & 1 \end{pmatrix}$. Descompunerea QR.

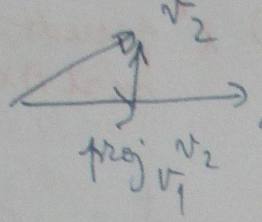
b) ? pseudosol pt sist $\begin{cases} x + 5y = 3. \\ 2x - 2y = 2. \\ -x + y = 5. \end{cases}$

a) $A=\begin{pmatrix}1&5\\2&-2\\-1&1\end{pmatrix}$. ? QR.

$v_1=\begin{pmatrix}1\\2\\-1\end{pmatrix}$; $v_2=\begin{pmatrix}5\\-2\\1\end{pmatrix}$; $\{v_1,v_2\}$ lin. ind.

v_1 v_2.

$\operatorname{rg}A=2 \Rightarrow v_1, v_2$ lin. ind. $\Rightarrow \{v_1,v_2\} \xrightarrow{G-S}$ $\{q_1,q_2\}$ sistem ortonormat.



$\begin{cases} u_1=v_1 \\ u_2=v_2-\text{proj}_{u_1} v_2 \end{cases}$ $u_1=(1,2,-1)^t$.

$u_2=\begin{pmatrix}5\\-2\\1\end{pmatrix}-\frac{\langle u_1,v_2\rangle}{\langle u_1,u_1\rangle}\cdot u_1$

$v_1 \perp v_2 \Rightarrow u_2=v_2$.

$q_1=\frac{v_1}{\|v_1\|}=\frac{1}{\sqrt{6}}\begin{pmatrix}1\\2\\-1\end{pmatrix}$; $q_2=\frac{v_2}{\|v_2\|}=\frac{1}{\sqrt{30}}\begin{pmatrix}5\\-2\\1\end{pmatrix}$.

$Q=[q_1|q_2]=\begin{pmatrix}1/\sqrt{6}&5/\sqrt{30}\\2/\sqrt{6}&-2/\sqrt{30}\\-1/\sqrt{6}&1/\sqrt{30}\end{pmatrix}$. $A=Q\cdot R$.

3x2 3x2 2x2

$R=\begin{pmatrix}r_{11}&r_{12}\\0&r_{22}\end{pmatrix}$ 21 22

$r_{11}=\langle v_1,q_1\rangle=\frac{1}{\sqrt{6}}[1+4+1]=\sqrt{6}$.

$r_{12}=\langle v_2,q_1\rangle=\frac{1}{\sqrt{6}}[5-4-1]=0$.

$r_{22}=\langle v_2,q_2\rangle=\frac{1}{\sqrt{30}}[5^2+2^2+1^2]=\sqrt{30}$.

$R=\begin{pmatrix}\sqrt{6}&0\\0&\sqrt{30}\end{pmatrix}$. $A=QR$.

b) $\begin{cases} x+5y=3 \\ 2x-2y=2 \\ -x+y=5 \end{cases}$ $A=\begin{pmatrix}1&5\\2&-2\\-1&1\end{pmatrix}$; $\vec{x}=\begin{pmatrix}x\\y\end{pmatrix}$; $\vec{b}=\begin{pmatrix}3\\2\\5\end{pmatrix}$.

$A\vec{x}=\vec{b}$; $A=QR \Rightarrow A^tA\vec{x}=A^t\vec{b} \Rightarrow Rx=Q^tb$.

$\begin{pmatrix}\sqrt{6}&0\\0&\sqrt{30}\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}1/\sqrt{6}&2/\sqrt{6}&-1/\sqrt{6}\\5/\sqrt{30}&-2/\sqrt{30}&1/\sqrt{30}\end{pmatrix}\begin{pmatrix}3\\2\\5\end{pmatrix} \Rightarrow x^*=\frac{1}{\sqrt{6}}\left[\frac{3+4-5}{\sqrt{6}}\right]=\frac{1}{3}$

$y^*=\frac{1}{\sqrt{30}}\cdot\frac{15-4+5}{\sqrt{30}}=\frac{16}{30}=\frac{8}{15}$

Ex 2 $\begin{cases} x+5y=3 \\ 2x-2y=2 \\ -x+y=5 \end{cases}$? pseudosol. exist.

$A=\begin{pmatrix} 1 & 5 \\ 2 & -2 \\ -1 & 1 \end{pmatrix}$; $b=\begin{pmatrix} 3 \\ 2 \\ 5 \end{pmatrix}$. $\boxed{Ax=b}$.

$A^tAx=A^tb$ sist compat.

$\boxed{x^*=(A^tA)^{-1}\cdot A^tb}$

$\underset{2\times3}{A^t}\cdot\underset{3\times2}{A}=\begin{pmatrix} 1 & 2 & -1 \\ 5 & -2 & 1 \end{pmatrix}\begin{pmatrix} 1 & 5 \\ 2 & -2 \\ -1 & 1 \end{pmatrix}=\begin{pmatrix} 6 & 0 \\ 0 & 30 \end{pmatrix}$.

$(A^t\cdot A)^{-1}=\frac{1}{180}\begin{pmatrix} 30 & 0 \\ 0 & 6 \end{pmatrix}=\begin{pmatrix} 1/6 & 0 \\ 0 & 1/30 \end{pmatrix}$.

$x^*=\begin{pmatrix} 1/6 & 0 \\ 0 & 1/30 \end{pmatrix}\cdot\begin{pmatrix} 1 & 2 & -1 \\ 5 & -2 & 1 \end{pmatrix}\begin{pmatrix} 3 \\ 2 \\ 5 \end{pmatrix}$.

Ex 3 a) $\{v_1=\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}; v_2=\begin{pmatrix} 5 \\ -2 \\ 1 \end{pmatrix}\}$; $b=\begin{pmatrix} 3 \\ 2 \\ 5 \end{pmatrix}$.

$\text{proj}_{Sp\{v_1,v_2\}}^{b}$

b) ? pseudosol pt. exist $\begin{cases} x+5y=3 \\ 2x-2y=2 \\ -x+y=5 \end{cases}$ incomp.

a) $\text{proj}_{Sp\{v_1,v_2\}}^{b} \overset{v_1\perp v_2}{=} \text{proj}_{v_1}^{b}+\text{proj}_{v_2}^{b}=\frac{\langle v_1,b\rangle}{\langle v_1,v_1\rangle}\cdot v_1+\frac{\langle v_2,b\rangle}{\langle v_2,v_2\rangle}\cdot v_2$.

$=\frac{2}{6}\cdot\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}+\frac{16}{30}\begin{pmatrix} 5 \\ -2 \\ 1 \end{pmatrix}=>$ (8, 15)

$\text{proj}_{Sp\{v_1,v_2\}}^{b}=\dots$.

b) $\boxed{Ax=\text{proj}_{Sp\{v_1,v_2\}}^{b}}$ sist. compatib. $\Rightarrow$ x^*.

Ex 4 $f\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}ax+y\\ax+by\end{pmatrix}$. i $f:\mathbb{R}^2\to\mathbb{R}^2$. liniară.

?. a,b ∈. f izomorfism și $\boxed{f^{-1}=-\frac{9}{2}f}$.

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f bij $\Leftrightarrow$ f inj $\Leftrightarrow \ker f=\{0_{\mathbb{R}^2}\}$

f surj $\Leftrightarrow \operatorname{Im} f=\mathbb{R}^2$.

$\ker f=\left\{\begin{pmatrix}x\\y\end{pmatrix} \middle/ \begin{pmatrix}ax+y\\ax+by\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix}\right\} \Rightarrow \begin{cases}ax+y=0\\ax+by=0\end{cases}$

~~$\Rightarrow y=-ax \Rightarrow ax(1-b)=0 \Rightarrow \boxed{b\neq 1}$ și $\boxed{a\neq 0}$.~~

• $\boxed{b=1} \Rightarrow f\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}ax+y\\ax+by\end{pmatrix}=x\begin{pmatrix}a\\a\end{pmatrix}+y\begin{pmatrix}1\\b\end{pmatrix}$.

$\operatorname{Im} f=\mathbb{R}^2 \Leftrightarrow \begin{pmatrix}a\\a\end{pmatrix};\begin{pmatrix}1\\b\end{pmatrix}$ liniar ind $\Leftrightarrow \operatorname{rg}\begin{pmatrix}a&1\\a&b\end{pmatrix}=2$.

$\Leftrightarrow ab-a\neq 0 \Rightarrow a(b-1)\neq 0 \Rightarrow \begin{cases}a\neq 0\\b\neq 1\end{cases}$.

$\boxed{M^{B_c}_{g\circ f}=M^{B_c}_{g}\cdot M^{B_c}_{f}}$.

f liniară $\Rightarrow f(\vec{v})=A\cdot\vec{v} \Rightarrow M^{B_c}_f=A$

$f\begin{pmatrix}x\\y\end{pmatrix}=\underbrace{\begin{pmatrix}a&1\\a&b\end{pmatrix}}_{A}\begin{pmatrix}x\\y\end{pmatrix}$.

$M^{-1}_{f^{-1}\circ f}=M^{-1}_{f}\cdot M_f$.

$f^{-1}\circ f=id \Rightarrow M^{B_c}_{id}=\begin{pmatrix}1&0\\0&1\end{pmatrix} \Rightarrow M_{f^{-1}}=(M_f)^{-1}$.

$f^{-1}\begin{pmatrix}x\\y\end{pmatrix}=A^{-1}\begin{pmatrix}x\\y\end{pmatrix}=\frac{1}{ab-a}\begin{pmatrix}b&-1\\-a&a\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}$.

$f^{-1}=-\frac{9}{2}f \Leftrightarrow \frac{1}{ab-a}\begin{pmatrix}b&-1\\-a&a\end{pmatrix}=\begin{pmatrix}a&1\\a&b\end{pmatrix}\cdot\left(-\frac{9}{2}\right)$.

$\frac{b}{ab-a}=-\frac{9}{2}a$ și $\frac{-1}{ab-a}=-\frac{9}{2}\mid\cdot a$ $\Rightarrow \boxed{b=-a}$ și $\frac{-a}{ab-a}=-\frac{9}{2}a$

$\Rightarrow 2=9a(b-1) \Rightarrow 2=-9a^2-9a$.

$9a^2+9a+2=0$.

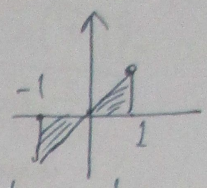
$\Delta=9^2-8\cdot 9=9$.

$a_{1,2}=\frac{-9\pm 3}{18}=\begin{cases}-1/3 \Rightarrow b=\frac{1}{3}\\-2/3 \Rightarrow b=2/3\end{cases}$.

Ex 5 $f: \mathbb{R}_{\le 2}[X] \to M_2(\mathbb{R})$. $f(P) = \begin{pmatrix} P(1)+P(-1) & P(1)-P(-1) \\ \int_{-1}^{1} xP(x)\,dx & 0 \end{pmatrix}$

M_f^{Bc}; $\operatorname{Im} f$; $\ker f$.

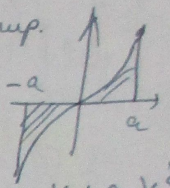
$f(1) = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$; $f(X) = \begin{pmatrix} 0 & 2 \\ \int_{-1}^{1} x^2 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ \frac{2}{3} & 0 \end{pmatrix}$.



$\int_{-1}^{1} x^2 = \int_{0}^{1} 2x^2 = \frac{2x^3}{3}\Big|_0^1$

$f(X^2) = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} \Rightarrow M_f^{Bc} = \begin{pmatrix} 2 & 0 & 2 \\ 0 & 2 & 0 \\ 0 & 2/3 & 0 \\ 0 & 0 & 0 \end{pmatrix}$.

$\int_{-1}^{1} x^3 = 0$ — fct imp.



$\operatorname{rg} M_f^{Bc} = 2 = \dim(\operatorname{Im} f)$.

$\ker f = \{ P \mid f(P) = 0 \}$. $P \in \mathbb{R}_{\le 2}[X]$; $P(X) = a_0 + a_1 X + a_2 X^2$.

$$\begin{cases} P(1)+P(-1)=0 \\ P(1)-P(-1)=0 \\ \int_{-1}^{1} xP(x)=0. \end{cases} \Leftrightarrow \begin{matrix} P(1)=0 \Rightarrow a_0+a_1+a_2=0. \\ P(-1)=0 \Rightarrow a_0-a_1+a_2=0 \end{matrix} \Leftrightarrow \begin{matrix} a_1=0 \\ a_2=-a_0 \end{matrix}$$

$P(X) = a_0 - a_0 X^2 = a_0(1-X^2)$.

$\int_{-1}^{1} X \cdot a_0(1-X^2)\,dx = a_0 \cdot \int_{-1}^{1} X - a_0 \cdot \int_{-1}^{1} X^3 = 0$. ✓

$\Rightarrow \ker f = \{ a_0(1-X^2) \mid a_0 \in \mathbb{R} \} = \operatorname{Sp}\{1-X^2\}$.

$\dim \ker f = 1$

$\operatorname{Im} f = \operatorname{Sp}\{ f(1), f(X), f(X^2) \} = \operatorname{Sp}\{ f(1), f(X) \}$

$= \operatorname{Sp}\left\{ \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 2 \\ 2/3 & 0 \end{pmatrix} \right\}$.

bază pt $\operatorname{Im} f$.

Ex 6 $A^2 - 3A + 2I_n = 0_2$; $A \in M_n(\mathbb{R})$ $\Rightarrow$? val proprii pt A

$A\vec{v} = \lambda \cdot \vec{v}$; $\vec{v}$ vector propriu.

$\lambda^2 - 3\lambda + 2 = 0$.

$\lambda_{1,2} = \dots$

Ex 7. $f: V \to V$ lin. ; $\dim V/\mathbb{R} = n$; $f^2 = f$. $\Rightarrow$ f diagonaliz.

$f(\vec{v}) = A \cdot \vec{v}$, $\forall \vec{v}$ $\Rightarrow$ $(f \circ f)(\vec{v}) = A^2 \cdot \vec{v}$. $\quad M_{f \circ f} = M_f \cdot M_f$.

$A^2 \cdot v = A \cdot v$. $\Rightarrow$ $(A^2 - A)\cdot \vec{v} = \vec{0}$; $\forall \vec{v}$.

v vect pr.

λ val pr. $\Rightarrow \lambda^2 - \lambda = 0$.

$A(A - I_n)\cdot \vec{v} = \vec{0}$, $\forall \vec{v}$.

$A = 0_n \Rightarrow f = 0_V$. $\Rightarrow$ diag

$A = I_n \Rightarrow f = id_V$ $\Rightarrow$ diag.

$A^2 - A = 0_n$.

$$\begin{cases} f \neq 0_V \\ f \neq 1_V \\ \det[A(A-I_n)] = 0 = \det A \cdot \det(A - I_n) \end{cases} \Rightarrow \begin{array}{l} \det A = 0 \Rightarrow 0 \in \sigma(A) \\ \det(A - I_n) = 0 \Rightarrow 1 \in \sigma(A) \end{array}$$

$\lambda_1 \quad \lambda_2$

$\sigma(A) = \{0, 1\}$.

f diag $\Leftrightarrow$ $V_{\lambda_1} \oplus V_{\lambda_2} = V$.

$\lambda_1 = 0 \Rightarrow V_{\lambda_1} = \{\vec{v} \mid f(\vec{v}) = \vec{0}\} = \ker f$.

$\lambda_2 = 1 \Rightarrow V_{\lambda_2} = \{\vec{v} \mid f(\vec{v}) = \vec{v}\} \subseteq \operatorname{Im} f$.

$V_{\lambda_2} \supseteq \operatorname{Im} f$.

$f^2 = f$.

$y \in \operatorname{Im} f \Rightarrow \ldots \Rightarrow y \in V_{\lambda_2}$.

$\exists x \in V$ i. $y = f(x)$. $\Rightarrow f(y) = f(f(x)) = f(x) = y$. $\Rightarrow y \in V_{\lambda_2}$ ☐

$V_{\lambda_1} = \ker f$ $\quad f: V \to V$ lin.

$V_{\lambda_2} = \operatorname{Im} f$. $\quad f^2 = f$.

$\ker f \oplus \operatorname{Im} f = V$. $\Rightarrow V_{\lambda_1} \oplus V_{\lambda_2} = V$. $\Rightarrow$

$\Rightarrow f$ diagonalizabil. ☐